

Definition and first terms

$(5 - 1\,x)\,A(x) = 4 + A(x)^4$, with the origin-normalized solution.
 Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+1*x*T+6*T^2+4*T^3+1*T^4$.
 $a(0),\dots,a(7) = 1, 1, 7, 95, 1614, 30718, 626434, 13383650$.
 Kernel used throughout: $\rho(u) = u\frac{1-6\,u^1-4\,u^2-1\,u^3}{1+1\,u}$, so $x = \rho(T(x))$.

Typogeometry

1x1

{1,1}x7

{1,{1,1}}x49

{1,1,1}x4

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 6, Δ_2^{marked} : 1, Δ_3 : 4, Δ_4 : 1.

Coefficient and contour extraction

$$\mathcal{K}_{n,h} = \{(k_1,\dots,k_3) \in \mathbb{Z}_{\geq 0}^3 : h + \sum_{j=1}^3 j k_j = n - 1\}, \quad K = \sum_{j=1}^3 k_j,$$

$$a(n) = \frac{1}{n} \sum_{h=0}^{n-1} \binom{n}{h} \sum_{\mathbf{k} \in \mathcal{K}_{n,h}} \binom{n+K-1}{K} \binom{K}{k_1,\dots,k_3} 6^{k_1} 4^{k_2} 1^{k_3}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$$\sum_{r=0}^4 P_r(n) a(n+r) = 0, \, n \geq 0, \text{ with}$$

$$P_0(n) = 3\,(n+1)\,(3\,n-1)\,(3\,n+7)$$

$$P_1(n) = -15\,(2\,n+3)\,(18\,n^2+54\,n+29)$$

$$P_2(n) = 75\,(n+2)\,(54\,n^2+216\,n+209)$$

$$P_3(n) = -6750\,(n+2)\,(n+3)\,(2\,n+5)$$

$$P_4(n) = 491\,(n+2)\,(n+3)\,(n+4)$$

Differential equation

The scalar linear ODE has order 3. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `release/attached_order4_certificate.json#/ode_from_recurrence`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$C(n,t) = \frac{(1+t)P(n,t)}{n(n+1)t^3Q(t)^3}.$$

Telescoping identity: $\sum_r P_r(n)H_{n+r}(u) = \partial_u(R(n,u)H_n(u))$.

Matrix dimensions: 8×8 ; remainder 3×4 , rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.